

GATE DA Practice Set: 4 Topics

Hypothesis Testing and CI | Bagging and Random Forest | FDs and Armstrong's Axioms | Probability Puzzles

What is in this PDF

Part A: where to find real GATE PYQs for each topic, with a tracking table. Part B: 40 new practice questions (10 per topic) with the answer and a short solution for each. Each question is tagged MCQ (one correct option), MSQ (one or more correct options) or NAT (numerical answer). The practice questions are original and written in GATE DA style; they are not past-paper questions.

Part A: Finding the real PYQs

GATE DA has had papers in 2024, 2025 and 2026, so each topic has only a few direct DA questions. I could not verify individual question numbers or years from the sources I reached, so I have not listed any. Use the official papers and answer keys, then fill the table below as you find questions.

- Official DA papers and keys for 2024 and 2025 are linked from the GeeksforGeeks page 'GATE CSE and DA Previous Years Papers' ([geeksforgeeks.org/gate/original-gate-previous-year-question-papers-cse-and-it-gq/](https://www.geeksforgeeks.org/gate/original-gate-previous-year-question-papers-cse-and-it-gq/)).
- A topic-wise bank for 2024, 2025 and 2026 is on AspirantMittraa (aspirantmittraa.com/exams/gate-da-pyq/).
- A community repository of DA resources is at github.com/DS-AI-GATE/dsai-gate.
- Hypothesis testing (z, t, chi-squared) is described by one source as added to the DA syllabus in 2026, so older DA papers may have few direct questions on it. Check the official syllabus.
- I could not confirm that bagging or random forest is named in the official DA syllabus. Check it before spending time on past papers for that topic.
- FDs, closure, keys and normal forms are asked often in GATE CS (DBMS) papers of earlier years, so those papers are a good source of extra PYQ practice for Topic 3.

Topic	Year	Paper	Q no.	Type	Solved?
Hypothesis testing / CI					
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Bagging / Random Forest					
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Bagging / Random Forest					
FDs / Armstrong					
FDs / Armstrong					
FDs / Armstrong					
Probability puzzles					
Probability puzzles					
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Send me any PYQs you collect and I will write step-by-step solutions in the same format.

Part B: New Practice Questions with Solutions

Topic 1: Hypothesis Testing and Confidence Intervals

Q1. [NAT] A random sample of size 36 is drawn from a normal population with known standard deviation 12. The sample mean is 50. Using $z = 1.96$, the lower limit of the 95% confidence interval for the population mean is _____ (rounded to 2 decimals).

Answer: 46.08 (accept 46.07 to 46.09)

Solution: Margin = $1.96 \times 12/\sqrt{36} = 1.96 \times 2 = 3.92$. Lower limit = $50 - 3.92 = 46.08$.

Q2. [MCQ] For a confidence interval for a population mean with known sigma, which change makes the interval WIDER, other things fixed?

- (A) Increasing the sample size
- (B) Decreasing the confidence level
- (C) Increasing the population standard deviation
- (D) None of the above

Answer: (C)

Solution: Width = $2 z \sigma/\sqrt{n}$. It grows with sigma and z (higher confidence) and shrinks with n.

Q3. [NAT] To test $H_0: \mu = 100$ against $H_1: \mu \neq 100$, a sample of size 25 from a normal population with known sigma = 15 gives sample mean 106. The value of the z statistic is _____.

Answer: 2

Solution: $z = (106 - 100)/(15/\sqrt{25}) = 6/3 = 2$.

Q4. [MCQ] In a two-sided z-test at significance level 0.01 (critical value 2.58), the computed statistic is $z = 2.4$. The correct decision is:

- (A) Reject H_0
- (B) Fail to reject H_0
- (C) The test is inconclusive
- (D) Accept H_1 with certainty

Answer: (B)

Solution: $|z| = 2.4$ is less than 2.58, so it lies in the acceptance region. Fail to reject H_0 (this is not proof that H_0 is true).

Q5. [NAT] A sample of 16 observations from a normal population has mean 52 and sample standard deviation 8. For $H_0: \mu = 50$, the t statistic is _____.

Answer: 1

Solution: $t = (52 - 50)/(8/\sqrt{16}) = 2/2 = 1$, with 15 degrees of freedom.

Q6. [MSQ] Which of the following statements are TRUE?

- (A) A Type I error is rejecting H_0 when H_0 is true.
- (B) Keeping n fixed, decreasing α increases the probability of a Type II error.
- (C) Keeping α fixed, increasing n decreases the probability of a Type II error.
- (D) The power of a test equals α .

Answer: (A), (B), (C)

Solution: A is the definition. For fixed n , α and β trade off (B). More data reduces β at fixed α (C). Power = $1 - \beta$, not α (D false).

Q7. [NAT] A sample of size 21 from a normal population has sample variance 30. To test $H_0: \sigma^2 = 20$, the chi-square test statistic is _____.

Answer: 30

Solution: Statistic = $(n-1) s^2 / \sigma_0^2 = 20 \times 30/20 = 30$, with 20 degrees of freedom.

Q8. [NAT] In a sample of 400 items, 100 are defective. The margin of error of the 95% ($z = 1.96$) confidence interval for the population proportion is _____ (rounded to 3 decimals).

Answer: 0.042 (accept 0.042 to 0.043)

Solution: $p\text{-hat} = 0.25$. $SE = \sqrt{0.25 \times 0.75/400} = \sqrt{0.00046875} = 0.02165$. Margin = $1.96 \times 0.02165 = 0.0424$.

Q9. [MSQ] A 95% confidence interval for μ computed from a sample is (48, 56). Which statements are TRUE?

- (A) The probability that μ lies in (48, 56) is 0.95.
- (B) In repeated sampling, about 95% of intervals constructed this way contain μ .
- (C) $H_0: \mu = 50$ is not rejected by the two-sided test at the 5% level.
- (D) $H_0: \mu = 60$ is rejected by the two-sided test at the 5% level.

Answer: (B), (C), (D)

Solution: Once computed, the interval either contains μ or not, so A is false. B is the frequentist meaning. By the duality of CIs and two-sided tests, values inside the interval (50) are not rejected and values outside (60) are rejected.

Q10. [NAT] The population standard deviation is 10. What is the smallest sample size that guarantees a 95% ($z = 1.96$) confidence interval for the mean with margin of error at most 2?

Answer: 97

Solution: $n \geq (z \sigma / E)^2 = (1.96 \times 10/2)^2 = 9.8^2 = 96.04$. Round up to 97.

Topic 2: Ensemble Methods - Bagging and Random Forest

Q1. [MCQ] A bootstrap sample of size n is drawn with replacement from n data points. For large n , the expected fraction of DISTINCT original points that appear in the bootstrap sample is approximately:

- (A) 36.8%
- (B) 50%
- (C) 63.2%
- (D) 100%

Answer: (C)

Solution: $P(\text{a given point is never picked}) = (1 - 1/n)^n \rightarrow 1/e = 0.368$. So about $1 - 0.368 = 0.632$ appear.

Q2. [NAT] A bootstrap sample of size 3 is drawn with replacement from 3 data points. The probability that a particular data point does NOT appear in the sample is _____ (rounded to 3 decimals).

Answer: 0.296 (accept 0.29 to 0.30)

Solution: $(2/3)^3 = 8/27 = 0.2963$.

Q3. [MCQ] The main benefit of bagging a high-variance learner such as a deep decision tree is a reduction in:

- (A) bias
- (B) variance
- (C) both bias and variance equally
- (D) irreducible error

Answer: (B)

Solution: Averaging many models trained on different bootstrap samples reduces variance. Bias stays roughly the same, and irreducible noise cannot be removed.

Q4. [NAT] A bagged regression ensemble has 5 trees that predict 10, 12, 11, 9 and 13 for a test point. The ensemble prediction is _____.

Answer: 11

Solution: Average = $(10 + 12 + 11 + 9 + 13)/5 = 55/5 = 11$.

Q5. [NAT] Three independent binary classifiers each have accuracy 0.7. A majority vote of the three is used. Assuming independent errors, the accuracy of the ensemble is _____ (rounded to 3 decimals).

Answer: 0.784 (accept 0.78 to 0.79)

Solution: At least 2 of 3 correct: $3 \times (0.7)^2 \times 0.3 + (0.7)^3 = 0.441 + 0.343 = 0.784$.

Q6. [MSQ] Which of the following statements about random forests are TRUE?

- (A) At each split only a random subset of the features is considered.
- (B) Individual trees are usually grown deep without pruning.
- (C) Feature subsampling makes the trees less correlated than plain bagged trees.
- (D) A random forest always has a strictly lower bias than a single fully grown tree.

Answer: (A), (B), (C)

Solution: A, B and C describe the standard algorithm. Random forests mainly cut variance; bias is about the same as a single deep tree, so D is false.

Q7. [NAT] Each of 25 independent estimators has variance 100. The variance of the average of the 25 estimators is _____.

Answer: 4

Solution: $\text{Var}(\text{mean}) = \sigma^2/B = 100/25 = 4.$

Q8. [MCQ] In bagging, the out-of-bag (OOB) samples for a tree are the training points not in its bootstrap sample. They are mainly used to:

- (A) estimate the generalization error without a separate validation set
- (B) increase the number of trees
- (C) prune each tree
- (D) reduce the number of features

Answer: (A)

Solution: Each point is predicted using only trees that did not see it, giving a nearly unbiased test-error estimate at no extra cost.

Q9. [NAT] A classification random forest uses the common default of $m = \sqrt{p}$ features considered at each split. If the dataset has $p = 16$ features, $m =$ _____.

Answer: 4

Solution: $\sqrt{16} = 4.$

Q10. [MSQ] Which of the following statements about bagging are TRUE?

- (A) The base models can be trained in parallel.
- (B) Increasing the number of bagged trees B generally leads to overfitting.
- (C) Bagging helps most for unstable, high-variance learners.
- (D) Bagging a stable linear regression usually gives a large improvement.

Answer: (A), (C)

Solution: Bootstrap models are independent, so they train in parallel (A). More trees do not typically overfit (B false). Stable learners gain little from bagging (D false).

Topic 3: Functional Dependencies and Armstrong's Axioms

Q1. [MCQ] Which of the following is NOT one of Armstrong's three basic axioms?

- (A) Reflexivity
- (B) Augmentation
- (C) Transitivity
- (D) Union

Answer: (D)

Solution: The axioms are reflexivity, augmentation and transitivity. Union is a derived rule.

Q2. [NAT] $R(A, B, C, D, E)$ has $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$. The number of candidate keys of R is _____.

Answer: 1

Solution: $A^+ = \{A, B, C, D, E\}$, so A is a key. No other attribute can be derived back to A , so no other single attribute or set without A can determine all. Only key: A .

Q3. [NAT] $R(A, B, C, D)$ has $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. The number of candidate keys is _____.

Answer: 3

Solution: B never appears on any right-hand side, so B is in every key. $AB^+ = ABCD$ ($AB \rightarrow C \rightarrow D$). $BC^+ = BCDA$ ($C \rightarrow D \rightarrow A$). $BD^+ = BDA$, then $AB \rightarrow C$ gives $ABCD$. No single attribute B works alone ($B^+ = B$). So the keys are AB , BC and BD : 3.

Q4. [MSQ] $R(A, B, C, D)$ has $F = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$. Which of the following FDs are implied by F ?

- (A) $A \rightarrow C$
- (B) $A \rightarrow BCD$
- (C) $B \rightarrow D$
- (D) $C \rightarrow A$

Answer: (A), (B)

Solution: $A^+ = \{A, B, C, D\}$, so $A \rightarrow C$ and $A \rightarrow BCD$ hold. $B^+ = \{B, C\}$ has no D . $C^+ = \{C\}$ has no A .

Q5. [NAT] $R(A, B, C, D, E, F)$ has $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. The number of attributes in the closure A^+ is _____.

Answer: 5

Solution: $A \rightarrow BC$ gives A, B, C . $B \rightarrow D$ adds D . $CD \rightarrow E$ adds E . $E \rightarrow A$ adds nothing new. $A^+ = \{A, B, C, D, E\}$. F is not included, so the size is 5.

Q6. [MCQ] $R(A, B, C)$ has $F = \{A \rightarrow B, B \rightarrow C\}$. The highest normal form R satisfies is:

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Answer: (B)

Solution: The only key is A . There are no partial dependencies because the key is a single attribute, so R is in 2NF. $B \rightarrow C$ is a transitive dependency (B is not a superkey and C is non-prime), so R is not in 3NF.

Q7. [NAT] The set $F = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$ is reduced to a minimal (canonical) cover. The number of FDs in the minimal cover is _____.

Answer: 2

Solution: Split $A \rightarrow BC$ into $A \rightarrow B$ and $A \rightarrow C$. $A \rightarrow C$ is redundant ($A \rightarrow B, B \rightarrow C$). $AB \rightarrow C$ is redundant since A is extraneous and $B \rightarrow C$ already exists. Minimal cover = $\{A \rightarrow B, B \rightarrow C\}$: 2 FDs.

Q8. [MSQ] Which of the following inference rules are VALID for functional dependencies?

- (A) If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$.
- (B) If $X \rightarrow YZ$, then $X \rightarrow Y$.
- (C) If $XY \rightarrow Z$, then $X \rightarrow Z$.
- (D) If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$.

Answer: (A), (B), (D)

Solution: A is union, B is decomposition, D is pseudotransitivity. C is invalid because Y may be needed to determine Z .

Q9. [MCQ] $R(A, B, C)$ has $F = \{A \rightarrow B\}$. Which decomposition is lossless-join?

- (A) $\{AB, BC\}$
- (B) $\{AB, AC\}$
- (C) $\{AC, BC\}$
- (D) $\{A, BC\}$

Answer: (B)

Solution: The common attribute of $\{AB, AC\}$ is A , and A is a key of AB ($A^+ = AB$), so the join is lossless. In the others the common attribute (B or C) determines neither relation, and in (D) there is no common attribute.

Q10. [NAT] $R(A, B, C)$ has $F = \{A \rightarrow BC\}$. The number of superkeys of R is _____.

Answer: 4

Solution: A is the only candidate key. Superkeys are A plus any subset of $\{B, C\}$: $2^2 = 4$.

Topic 4: Probability - Aptitude and Puzzles

Q1. [NAT] Two fair dice are rolled. Given that at least one die shows 3, the probability that the sum is 7 is _____ (rounded to 2 decimals).

Answer: 0.18 (accept 0.18 to 0.19)

Solution: Outcomes with at least one 3: 11. Sum 7 with a 3: (3,4) and (4,3) = 2. Probability = $2/11 = 0.1818$.

Q2. [MCQ] A family has two children and at least one is a boy. Assuming boys and girls are equally likely and independent, the probability that both are boys is:

(A) $1/4$

(B) $1/3$

(C) $1/2$

(D) $2/3$

Answer: (B)

Solution: Sample space given at least one boy: {BB, BG, GB}. Probability of BB = $1/3$.

Q3. [NAT] Monty Hall: you pick one of three doors, the host (who knows where the car is) opens a different door showing a goat, and you switch to the remaining door. The probability of winning the car is _____ (rounded to 2 decimals).

Answer: 0.67 (accept 0.66 to 0.67)

Solution: Switching wins exactly when your first pick was a goat, which has probability $2/3$.

Q4. [NAT] An urn has 5 red and 3 blue balls. Two balls are drawn without replacement. The probability that both are the same colour is _____ (rounded to 2 decimals).

Answer: 0.46 (accept 0.46 to 0.47)

Solution: $(C(5,2) + C(3,2))/C(8,2) = (10 + 3)/28 = 13/28 = 0.464$.

Q5. [MCQ] 1% of a population has a disease. A test is positive for 90% of diseased people and also for 10% of healthy people. A person tests positive. The probability that the person has the disease is approximately:

(A) 0.083

(B) 0.5

(C) 0.9

(D) 0.1

Answer: (A)

Solution: $P(D \text{ and } +) = 0.01 \times 0.9 = 0.009$. $P(+) = 0.009 + 0.99 \times 0.1 = 0.108$. $P(D | +) = 0.009/0.108 = 1/12 = 0.083$.

Q6. [NAT] A fair coin is tossed repeatedly. The expected number of tosses needed to get two consecutive heads for the first time is _____.

Answer: 6

Solution: $E = 1/2 (1 + E) + 1/4 (2 + E) + 1/4 (2)$. So $E = 0.5 + 0.5E + 0.5 + 0.25E + 0.5$, giving $0.25E = 1.5$ and $E = 6$.

Q7. [NAT] Three people are each born on a uniformly random day of the week, independently. The probability that at least two of them were born on the same day is _____ (rounded to 2 decimals).

Answer: 0.39 (accept 0.38 to 0.39)

Solution: $P(\text{all different}) = (7 \times 6 \times 5)/7^3 = 210/343 = 30/49$. Required = $19/49 = 0.388$.

Q8. [MSQ] For events A and B, $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$. Which statements are TRUE?

- (A) $P(A \text{ and } B) = 0.2$
- (B) A and B are independent
- (C) A and B are mutually exclusive
- (D) $P(A | B) = 0.5$

Answer: (A), (B), (D)

Solution: $P(A \text{ and } B) = 0.5 + 0.4 - 0.7 = 0.2$. Since $0.5 \times 0.4 = 0.2$, they are independent. They are not mutually exclusive since the intersection is nonzero. $P(A | B) = 0.2/0.4 = 0.5$.

Q9. [NAT] Two players alternately toss a coin with $P(\text{head}) = 1/3$. The first to get a head wins, and player 1 tosses first. The probability that player 1 wins is _____.

Answer: 0.6

Solution: $P = p + (1-p)^2 p + (1-p)^4 p + \dots = p/(1 - (1-p)^2) = (1/3)/(1 - 4/9) = (1/3)/(5/9) = 3/5$.

Q10. [MCQ] Three cards: one is red on both sides, one is blue on both sides, and one is red on one side and blue on the other. A card is drawn at random and a random side is shown: it is red. The probability that the other side is also red is:

- (A) $1/2$
- (B) $2/3$
- (C) $1/3$
- (D) $3/4$

Answer: (B)

Solution: There are 3 equally likely red faces: two on the red-red card and one on the mixed card. In 2 of the 3 the other side is red, so $2/3$.